

Exam Guidelines – Data Management and Analysis

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Overview

Each student or group (maximum of **four** members) will present a **7-minute** data analysis project during the oral exam. The aim is to demonstrate a **rigorous data science pipeline**: from dataset selection and cleaning, through exploratory and inferential analysis, to the extraction of meaningful insights.

This is not a showcase of code or design. It is a scientific presentation — grounded in the techniques and principles discussed throughout the course.

The only required **output** is a set of **slides**, presented during the exam. No prior submission is needed.

Project Requirements

Dataset

- Must be **publicly accessible**, well-structured, and properly documented.
- Should allow for **non-trivial**, quantitative investigation.
- All preprocessing and transformation steps must be clearly explained and justified.

Research Question

- Must be **specific**, **relevant**, and statistically approachable.
- Avoid vague or purely descriptive goals.
- Explain why the question matters and what insights are expected.

Methodology

- Apply techniques covered in the course: data preprocessing, grouping, aggregation, visualization, hypothesis testing, linear modeling.
- **Use only methods you understand.**
- All results must be reproducible and assumptions clearly stated.

Presentation

- Slides should be clean, readable, and focused.
- Use visuals only if they support the reasoning.
- Follow a clear narrative: *motivation* → *methods* → *results* → *interpretation*.

Evaluation Criteria

1. **Data Management:** Quality of preprocessing, filtering, and feature engineering.
2. **Analytical Rigor:** Appropriate use of techniques; reproducibility; validity of interpretation.
3. **Communication:** Logical flow, slide quality, and capacity to answer questions precisely.

What Invalidates the Exam

The following behaviors will compromise or nullify your exam:

- **Copying** work from other students or prior cohorts.
- **Using ChatGPT or AI tools without explicit declaration.**
- Employing tools/libraries you do not fully understand.
- Misapplying statistical methods (e.g., using linear regression without checking assumptions).
- Drawing conclusions not supported by the data.
- Using visualizations as decorative elements without real analysis.
- Obscuring methodology or fabricating results.
- Overfitting or cherry-picking metrics (e.g., accuracy without context).
- Using black-box notebooks found online without understanding.
- Ignoring outliers, nulls, or key data irregularities.
- Hand-waving explanations or inability to justify choices.
- **Confusing storytelling with analysis:** your goal is to answer a question, not to tell a feel-good narrative.

Tips for Success

- Choose a dataset that raises a real question—not just one that is “cool.”
- Stay within the tools discussed in class or study any extensions deeply.
- Interpret what the data says—and what it doesn’t.
- Avoid filler slides: each one must carry weight.
- Practice to ensure your delivery fits the 7-minute limit.
- **Seems easy? It isn’t. But if you do it well, it shows. And it’s worth it.**